Car Price Prediction using Machine Learning

Objective

The objective of this project is to predict the selling price of cars using machine learning techniques.

Dataset

The dataset contains:

- Car Name
- Year
- Present Price
- Driven Kilometers
- Fuel Type
- Selling Type
- Transmission
- Owner Details
- Selling Price

Technologies Used

- Python
- Pandas
- Scikit-learn
- Random Forest Regressor

Methodology

1. Load and preprocess the dataset.
2. Create a new feature called Car Age.
3. Encode categorical variables.
4. Split data into training and testing sets.
5. Train a Random Forest Regression model.
6. Evaluate model performance using MAE and R^2 Score.

Results

The model achieved:

- MAE = 0.6364
- R^2 Score = 0.9599

Present Price was identified as the most important factor affecting car selling prices.

Conclusion

The project demonstrates how machine learning can be used to accurately predict car prices and assist in automobile valuation.